



Analysis of Soluble α -Synuclein Aggregates in CSF by Semi-Quantitative Seed Amplification Assay (SAA)

John Middleton, Karen MacLeod
Amprion Inc. Clinical Laboratory

PPMI Project ID: 207

Summary

The goal of Project 207 is to demonstrate feasibility of limiting dilution as the basis for a semi-quantitative seed amplification assay (SAA) for analysis of soluble α -synuclein (α Syn) aggregates in cerebrospinal fluid (CSF). Samples from the PPMI repository unintentionally exposed to full or partial thawing are being used since SAA, unlike other biomarker tests, has been shown to be robust against sample freeze/thaw. In project Steps 1 and 2A, serial dilutions of individual PD CSF samples were performed using a pool of control CSF to develop a dilution strategy that captures variability between samples. In Step 2B, this dilution strategy was applied for semi-quantitative analysis of longitudinal PD CSF samples to evaluate whether a change in the level of α synuclein aggregates is seen with progressing disease. For each step, undiluted samples were initially analyzed under blinded conditions using Amprion's validated qualitative SAA assay (SYNTap[®] Biomarker Test); results from blinded analysis of undiluted samples were used to assess accuracy of the qualitative assay in this sample cohort. Semi-quantitative analysis was performed in Amprion's Clinical Laboratory using an RUO version of the test incorporating modifications to accommodate sample dilution and available sample volume (refer to Method Section). Results are provided in a staged fashion after each project step.

Method

To perform the SYNTap[®] Biomarker Test, each CSF sample is analyzed in triplicate. For each replicate, a borosilicate glass bead, 40 μ L of CSF, and 160 μ L of reaction buffer are combined in a single well of a 96-well plate. Reaction buffer is formulated such that the final incubation mixture contains 0.3 mg/mL recombinant full-length α Syn (containing his-tag for purification), 100 mM PIPES buffer pH 6.5, 500 mM NaCl, and 10 μ M Thioflavin T (ThT). For each plate, QC is performed by testing Amprion-formulated negative and positive assay controls. Plates are sealed with optical adhesive film and incubated at 37°C while being subjected to cyclic agitation (one minute shaking followed by 29 minutes without shaking). Fluorescence is read at the start of the test and once daily for 7-10 days using a FLUOStar Ω Microplate Reader (excitation wavelength 440 nm; emission wavelength 490 nm). Samples with a median maximum fluorescence equal to or greater than a cutoff of 25,000 RFU at the end of the incubation period are classified as "Detected"; samples with median maximum fluorescence values below the 25,000 RFU cutoff are classified as "Not Detected".





The RUO version of the test used for the Step 2B semi-quantitative analysis was performed as above with the following modifications: reaction volume 20 μ l; 0.1% Sarkosyl included in the incubation mixture; and fluorescence read twice a day for three days. For this step, the reported Spearman-Kärber SD50 estimates are based on the hit rate versus dilution data contained in Appendix A.

Anticipated Data Return Timeline

Step 1 Prescreening: June 2021
Step 1 Dilution Data: August 2021
Step 2A Prescreening: July 2021
Step 2A Dilution Data: September 2021
Step 2B Prescreening: September 2021
Step 2B Dilution Data: June 2022

References

N/A

About the Authors

This document was prepared by Karen MacLeod and John Middleton, Amprion Inc. For more information, please contact Karen MacLeod by email at Karen@AmprionDx.com.

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Appendix A – Data and Methods used for SD50 Determinations

The Spearman-Kärber SD50 calculations were performed using the RStudio TSK() function. The function call used is shown below. Refer to the 'ecotoxicology' Package for additional information.

Function - TSK(x, r, n, A = 0.49, conf = 0.95)

Arguments

x numeric vector – vector of dilution values

r numeric vector – vector of hit counts where a hit is defined as a reaction well with FMAX data that exceeds the defined 25,000 RFU threshold

n numeric vector – vector of trials at each dilution, n=5 for each dilution tested

A numeric vector – trimming variable set to 0.49

conf numeric vector – the default is 0.95

<u>PATNO</u>	<u>PLATE</u>	<u>RUNDATE</u>	<u>SAMPLE</u>	HITS BY DILUTION							<u>SD50</u>
				<u>1.000</u>	<u>0.667</u>	<u>0.333</u>	<u>0.111</u>	<u>0.037</u>	<u>0.012</u>	<u>0.000</u>	
4096	SAA1232	5/17/2022	PP0029-1246	5	4	3	5	2	1	0	0.049
4096	SAA1232	5/17/2022	PP0034-0245	4	5	2	1	0	0	0	0.383
3500	SAA1233*	5/17/2022	PP0029-3511	5	5	5	3	1	1	0	0.045
3500	SAA1233*	5/17/2022	PP0034-8322	5	5	4	3	0	1	0	0.048
3052	SAA1234*	5/17/2022	PP0028-4923	5	5	5	4	2	2	0	0.023
3052	SAA1234	5/17/2022	PP0029-3327	5	5	5	5	2	0	0	0.044
3024	SAA1236	5/23/2022	PP0028-9017	5	5	5	2	3	0	0	0.064
3024	SAA1236	5/23/2022	PP0029-1968	5	4	5	4	0	0	0	0.074
3024	SAA1236	5/23/2022	PP0034-0285	5	5	5	4	0	0	0	0.074
3051	SAA1237	5/23/2022	PP0029-3489	5	5	4	5	2	0	0	0.046
3051	SAA1237	5/23/2022	PP0029-3869	5	5	5	4	1	1	0	0.064





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3051	SAA1237	5/23/2022	PP0034-7082	5	5	5	4	1	1	0	0.064
3067	SAA1238	5/23/2022	PP0028-5379	5	5	5	2	1	1	0	0.133
3067	SAA1238	5/23/2022	PP0029-2483	5	5	5	4	1	0	0	0.064
3067	SAA1238	5/23/2022	PP0029-3923	5	5	5	2	1	1	0	0.133
3078	SAA1239	5/23/2022	PP0028-8776	5	5	5	4	2	0	0	0.049
3078	SAA1239	5/23/2022	PP0029-2241	5	5	4	4	3	0	0	0.031
3078	SAA1239	5/23/2022	PP0034-7096	5	5	5	5	0	1	0	0.049
3102	SAA1240	5/24/2022	PP0029-3459	5	5	5	5	3	1	0	0.028
3102	SAA1240**	5/24/2022	PP0029-3961	5	5	5	5	4	3	0	0.010
3102	SAA1240	5/24/2022	PP0034-8319	5	5	5	4	2	0	0	0.049
3126	SAA1241	5/24/2022	PP0028-8116	5	5	5	4	1	0	0	0.064
3126	SAA1241	5/24/2022	PP0029-1990	5	5	4	1	2	0	0	0.172
3126	SAA1241	5/24/2022	PP0034-0239	5	5	5	3	1	3	0	0.064
3229	SAA1242	5/24/2022	PP0028-8106	5	4	3	2	0	0	0	0.192
3229	SAA1242	5/24/2022	PP0029-2154	5	5	5	3	0	1	0	0.089
3229	SAA1242	5/24/2022	PP0034-1902	5	4	2	3	1	0	0	0.192
3307	SAA1243	5/24/2022	PP0029-0414	5	5	4	2	1	0	0	0.146
3307	SAA1243	5/24/2022	PP0034-5714	5	5	5	1	1	0	0	0.168
3307	SAA1243	5/24/2022	PP0034-9823	5	5	4	3	0	0	0	0.092
3359	SAA1248	5/31/2022	PP0029-0299	5	5	5	2	1	1	0	0.133
3359	SAA1248	5/31/2022	PP0029-3896	5	5	5	1	2	0	0	0.152
3359	SAA1248	5/31/2022	PP0033-6810	5	5	5	4	1	2	0	0.057
3373	SAA1249	5/31/2022	PP0028-8992	5	4	3	2	0	1	0	0.192
3373	SAA1249	5/31/2022	PP0029-0486	5	5	3	3	1	0	0	0.084
3373	SAA1249	5/31/2022	PP0029-4223	5	5	4	1	0	0	0	0.192
3377	SAA1250	5/31/2022	PP0028-9021	5	5	5	1	0	0	0	0.168
3377	SAA1250	5/31/2022	PP0029-2214	5	5	5	2	0	2	0	0.133
3377	SAA1250	5/31/2022	PP0029-4468	5	5	4	3	4	0	0	0.027

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3403	SAA1251	5/31/2022	PP0029-2957	5	5	5	5	2	2	0	0.044
3403	SAA1251	5/31/2022	PP0029-3465	5	5	5	2	0	2	0	0.133
3403	SAA1251	5/31/2022	PP0029-3532	5	5	5	4	3	0	0	0.031
3407	SAA1255	6/6/2022	PP0029-3479	5	5	5	3	2	1	0	0.064
3407	SAA1255	6/6/2022	PP0029-3546	5	5	5	3	3	1	0	0.028
3407	SAA1255	6/6/2022	PP0034-0269	5	5	5	4	2	0	0	0.049
3409	SAA1256	6/6/2022	PP0029-3482	5	5	5	4	1	0	0	0.064
3409	SAA1256	6/6/2022	PP0029-3550	5	5	5	4	3	1	0	0.028
3409	SAA1256	6/6/2022	PP0034-5695	5	5	5	2	1	0	0	0.133
3418	SAA1257	6/6/2022	PP0029-0226	5	5	3	0	1	0	0	0.267
3418	SAA1257	6/6/2022	PP0029-3910	4	5	3	2	0	1	0	0.192
3418	SAA1257	6/6/2022	PP0033-7396	5	5	4	3	0	1	0	0.089
3654	SAA1258	6/6/2022	PP0028-6730	4	5	1	3	1	0	0	0.383
3654	SAA1258	6/6/2022	PP0029-3886	3	3	2	0	0	0	0	0.471
3654	SAA1258	6/6/2022	PP0033-7387	5	4	3	0	1	1	0	0.263
4001	SAA1259	6/7/2022	PP0029-0016	5	5	4	4	1	1	0	0.064
4001	SAA1259	6/7/2022	PP0029-2386	5	5	5	4	1	1	0	0.064
4001	SAA1259	6/7/2022	PP0034-5855	5	5	4	2	3	2	0	0.063
4024	SAA1260	6/7/2022	PP0028-8486	5	5	5	5	3	0	0	0.031
4024	SAA1260	6/7/2022	PP0029-2160	5	5	5	4	3	0	0	0.031
4024	SAA1260	6/7/2022	PP0033-6259	5	5	5	3	3	0	0	0.031
4027	SAA1261	6/7/2022	PP0028-7336	0	1	0	0	0	0	0	not detected
4027	SAA1261	6/7/2022	PP0029-2276	0	0	0	0	0	1	0	not detected
4027	SAA1261	6/7/2022	PP0034-0226	1	0	0	0	1	0	0	not detected
4071	SAA1262	6/7/2022	PP0028-9562	0	0	0	0	1	0	0	not detected
4071	SAA1262	6/7/2022	PP0029-1159	1	0	0	0	0	0	0	not detected
4071	SAA1262	6/7/2022	PP0033-6231	0	2	0	0	1	0	0	not detected
4078	SAA1264	6/13/2022	PP0028-9571	0	1	0	1	0	1	0	not detected

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4078	SAA1264	6/13/2022	PP0033-6238	0	0	0	0	0	0	0	not detected
4078	SAA1264	6/13/2022	PP0034-7072	1	0	0	0	0	0	0	not detected
4091	SAA1265	6/13/2022	PP0028-7273	5	4	5	2	2	0	0	0.138
4091	SAA1265	6/13/2022	PP0029-2607	5	5	5	3	1	0	0	0.084
4091	SAA1265	6/13/2022	PP0029-4443	5	5	5	3	0	1	0	0.089
3478	SAA1266	6/14/2022	PP0028-4994	5	5	5	5	3	2	0	0.021
3478	SAA1266	6/14/2022	PP0028-4995	5	5	5	5	4	2	0	0.016

* - 1.000 0.400 0.250 0.063 0.016 0.004 0.000 (different dilution scheme used)

** - SD50 not calculable, estimate based on linear interpolation

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